

Assignment 6 – Project Prototype

Similar Products, UX Decisions, and Feature Scope

Similar Products Researched

iClicker (Classroom Response & Attendance System)

iClicker is a classroom response and attendance system that allows instructors to track student participation and attendance using a mobile app or physical clicker devices.

Interface Elements Borrowed:

- Session-based attendance views
- Real-time updates during class sessions
- Clear present/absent status indicators
- Instructor-focused dashboards

Interface Limitations:

- Largely single-purpose, focused only on attendance and polling
- Requires students to install and log into a dedicated app
- Limited flexibility for labs, workshops, or non-traditional environments

How This Project Improves Upon It: This project treats attendance as one configurable task within a larger tap-based system, allowing the same interaction model to support additional use cases.

Blackboard Attendance (LMS Feature)

Blackboard Attendance is an LMS-based feature that allows instructors to manually record attendance using official course rosters.

Interface Elements Borrowed:

- Roster-based student verification
- Clear association between students and courses
- Familiar instructor workflows

Interface Limitations:

- Attendance is fully manual and time-consuming
- No physical interaction or automation
- Limited real-time feedback during class

How This Project Improves Upon It: The proposed system automates attendance using intentional NFC taps while maintaining roster verification.

QR Code–Based Attendance Systems

QR-based systems allow students to scan a displayed code to register attendance.

Interface Limitations:

- Easily exploited through remote scanning or screenshots
- Weak identity verification
- Heavy reliance on personal devices

How This Project Improves Upon It: NFC-based interactions require physical presence, improving trust while maintaining ease of use.

Prioritized Feature List (Final Product Vision)

- 1 Task-based NFC tap interaction system
- 2 Support for NFC cards and mobile devices
- 3 Configurable task types (attendance, events, access, automation)
- 4 Session-based rule configuration
- 5 Real-time feedback and verification
- 6 Instructor and administrator dashboards

Features Supported by the Prototype

- Instructor dashboard with active session overview
- Live attendance list with student status indicators
- NFC tap representation for student check-ins
- Hover or tap-based student profile previews
- Configurable session rules and active task concept

References / Products Reviewed

- iClicker. Classroom response and attendance system. <https://www.iclicker.com>
- Blackboard. Blackboard attendance feature. <https://www.blackboard.com>

- Various QR-based attendance tools used in higher education environments.
- RFID/NFC campus identification and access systems used by universities.